

Madhvika Sehgal

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SUMMARY

Business Intelligence Engineer with 4+ years of experience in analytics, finance, and operational data solutions across fintech and healthcare-tech. Proficient in SQL pipelines, cost allocation frameworks, and KPI reporting using AWS, Tableau, Python, and cloud data platforms. At Amazon, built Contribution Profit and MEC cost allocation pipelines for Pharmacy Finance reporting through work-unit derivation, cost attribution, and month-end reconciliation. At Barclays, led Call Center Analytics driving SLA optimization and reporting automation. Skilled at translating ambiguous business problems into scalable solutions through cross-functional collaboration.

KEY SKILLS

Programming & Data Engineering: SQL, Python (Pandas, NumPy), SAS, Data Modeling, Git, Data Transformation and Validation

Cloud & Data Platforms: AWS (S3, Athena, Glue, Lambda, Redshift), Oracle, Tableau, Tableau Prep, Alteryx

Analytics Engineering & BI: KPI Development, Funnel Analysis, Customer Journey Analytics, ETL Processes, Finance Analytics, Activity-Based Costing (ABC), Operational Cost Attribution

Project Management: Agile, JIRA, Confluence, Kanban, Cross-Functional Stakeholder Management

WORK EXPERIENCE

Business Intelligence Engineer

Amazon (Aug 2025 – Present)

- **SME:** Led analysis, debugging, enhancement, and validation of a 1,900+ line SQL-based **cost allocation** pipeline, supporting allocation of fulfillment, customer service, pharmacist, and operational costs into **Pharmacy Contribution Profit** reporting.
- **Cost Allocation Redesign:** Redesigned allocation logic during **cost center restructuring**, validating downstream impacts through structured allocation models and SQL reconciliation to ensure accurate financial attribution.
- **GenAI-Assisted Cost Analysis:** Integrated Claude via Amazon Bedrock to analyze cost allocation anomalies and generate natural language summaries of month-end variance reports, reducing manual interpretation time and improving stakeholder communication of complex financial data.
- **Automated Labor Allocation:** Built an automated **labor allocation solution** supporting Pharmacy month-end reporting by connecting **ADP payroll data, warehouse metadata, and employee-to-warehouse allocation logic**, enabling dynamic onboarding of new fulfillment centers, eliminating manual S3 file updates, and saving **4+ hours** of recurring monthly effort.
- **Data Platform Usage Metrics Dashboard:** Built a governance dashboard using **audit log event data (1M+ monthly events)** to measure dataset adoption, platform usage, user access patterns, and unused assets, enabling better data platform governance and capacity planning.
- **Business-Facing Documentation:** Authored a comprehensive business-facing framework document explaining **cost allocation methodology, cost drivers, manual adjustments, and rate-card logic**, simplifying highly technical SQL and finance workflows for stakeholder consumption.

Business Intelligence Analyst

Barclays (Dec 2020 – Jul 2025)

- **Callback Feature Revamp:** Led a 3-member team to redesign SLA logic and integrate callback features into Contact Center workflows. Enhanced 45+ dashboards and reports boosting queue efficiency & improving customer retention by 15%.
- **GenAI-Driven Automation:** Engineered a serverless solution using **AWS Lambda and Claude LLM via Bedrock** to automate report metadata summarization, significantly improving audit readiness and version control from days to minutes.
- **Cloud Migration & Data Lake Setup:** Co-led the migration of multi-terabyte datasets from **Oracle to AWS S3**, creating a scalable data lake that cut query latency by 40% using **Athena and Redshift**.
- **Customer Journey Analytics:** Analyzed clickstream, call log, and servicing data across 5M+ IVR interactions. Created heatmaps and predictive **Tableau** models that helped reduce call time by 15% and transfers by 20%.
- **ETL Framework for IVR Dashboards:** Built and managed end-to-end **ETL processes** powering over 30 IVR **Tableau dashboards**, enabling standardized data pipelines for real-time funnel monitoring and KPI consistency.
- **A/B Testing for Queue Optimization:** Led design and execution of **A/B tests for queue restructuring** (Simple vs Complex), boosting resolution efficiency by 12%, and developed a Tableau dashboard to track month-over-month performance.
- **Agent Performance Scoring Model:** Developed a scoring model using sentiment, CSAT, and NPS data, incorporating **Z-score normalization** to benchmark agent performance and identify outliers for QA and coaching.
- **Real-Time Marketing Insights:** Developed Redshift-powered dashboards supporting real-time **KPI tracking**, leading to a 20% increase in IVR containment and reducing manual reporting effort by 80%.

- **IVR Funnel Monitoring:** Built comprehensive dashboards for funnel KPIs like containment, self-service, and sentiment; introduced tagging improvements that cut call transfers by 30%.
- **Cross-Functional Enablement:** Delivered scalable analytics pipelines for product, Customer Experience (CX), Fraud- Disputes, and Collections teams. Used **Agile (JIRA/Confluence)** to manage workflows, increasing delivery speed by 30%.

Business Consultant

Accenture (Sep 2016 – Jul 2019)

- Led global SAP MM implementations and upgrades for global clients, delivering **full-cycle project management** from requirements gathering to deployment, using **Agile** methodologies. Specialized in Procure-to-Pay (P2P) across purchasing, inventory & invoice.
- Streamlined procurement operations by implementing workflow **automation**, including automatic PO generation and **release strategy configuration** resulting in a 25% reduction in processing time & improved operational efficiency.
- Facilitated cross-functional collaboration between business and IT teams to ensure **seamless integration** of MM with SD, FI, and PP modules, enhancing data integrity, inter-system communication, and **enterprise resource planning** continuity.
- Developed functional specifications and **business requirement documents (BRDs)** for custom reports and enhancements; partnered with ABAP developers to deliver scalable, logic-driven solutions aligned with business **KPIs and change management**.
- Conducted stakeholder workshops, **end-user training, and user acceptance testing (UAT)**; provided **post-go-live support** that resulted in 95% user adoption, high stakeholder satisfaction, and minimal disruption to business operations.

EDUCATION

Master of Science, Management Information Systems, Focus – Business Intelligence and Data Analytics

University of Arizona, Tucson, AZ (Aug 2019 – Dec 2020)

Bachelor of Technology, Information Technology

Guru Gobind Singh Indraprastha University, New Delhi, India (August 2012- May 2016)

CERTIFICATIONS

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- Gen AI Product Management Specialization, Duke University (Coursera, 2025)
 - Generative AI: Prompt Engineering Basics, IBM (Coursera, 2025)